

Practice Program-4

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Q 1. Create a class 'Student' with rollno, studentName, dictionary of marks(subjectName -> marks [3]).

Provide following functionalities

A. initializer

B. override __str__ method

C. Print student data for given id.

D. Calculate GPA()

$$\text{gpa} = (1/3) * m1 + (1/2) * m2 + (1/4) * m3$$

Create 5 student objects and store them in a list

For the student data stored in the list perform following operation:

1. Display All Student
2. Search by id
3. Sort by name
4. Calculate GPA of a student

Program:-

```
class Student:

    #initializer(__init__)
    def __init__(self, rollno, studentName, marks):
        self.rollno=rollno
        self.studentName=studentName
        self.marks=marks

    #implement __str__ method
    def __str__(self):
        return f"{self.rollno}: {self.studentName}
{self.marks}"

    #Calculate GPA()
    def gpa(self):
```

```

        gpa = (1 / 3) * self.marks['Maths'] + (1 / 2) *
self.marks['Science'] + (1 / 4) * self.marks['English']
        return gpa

s1= Student(11,'Ashlesha', {'Maths':60,'Science':
88,'English': 92})
s2= Student(12,'Ruchika', {'Maths':77,'Science':
87,'English': 86})
s3= Student(13,'Payal', {'Maths':56,'Science': 78,'English':
94})
s4= Student(14,'Ajay', {'Maths':50,'Science': 79,'English':
70})
s5= Student(15,'Pranita', {'Maths':85,'Science':
88,'English': 72})
# print(s1.get_student_by_id())
student=[s1,s2,s3,s4,s5]

print(f"{'RollNo':<10}{'Name':<19}{'Marks':<12}{'GPA'}")
for s in student:

print(f"{s.rollno:<10}{s.studentName:<14}{s.marks['Maths']:<
4} {s.marks['Science']:<4}
{s.marks['English']:<6}{round(s.gpa(),2)}")

print(end='\n')
n=int(input('Enter student rollno: '))

for s in student:
    if s.rollno==n:

print(f"{s.rollno:<10}{s.studentName:<14}{s.marks['Maths']:<
4} {s.marks['Science']:<4}
{s.marks['English']:<6}{round(s.gpa(), 2)}")

print(end='\n')
#Sort by name
print("Sorted Data")
sorted_data = sorted(student, key=lambda s: s.studentName)

for s in sorted_data:

```

```
print(f"{s.rollno:<10}{s.studentName:<14}{s.marks['Maths']:<4} {s.marks['Science']:<4} {s.marks['English']:<6}{round(s.gpa(),2)}")
```

Output:-

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day5\Student.py
RollNo    Name           Marks          GPA
11        Ashlesha       60  88  92    87.0
12        Ruchika        77  87  86    90.67
13        Payal          56  78  94    81.17
14        Ajay           50  79  70    73.67
15        Pranita        85  88  72    90.33

Enter student rollno: 13
13        Payal          56  78  94    81.17

Sorted Data
14        Ajay           50  79  70    73.67
11        Ashlesha       60  88  92    87.0
13        Payal          56  78  94    81.17
15        Pranita        85  88  72    90.33
12        Ruchika        77  87  86    90.67

Process finished with exit code 0
```